

Upayan Roy Chowdhury

AI Software Engineer · Astrodynamics · Founder, DragonOS

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PROFESSIONAL SUMMARY

Astrodynamics and AI Software Engineer with a B.Tech in Aerospace Engineering (SRM University) and an MSc in Astronautics & Space Engineering (Cranfield University). 1.5 years in aerospace engineering and 1.5+ years building production AI systems — RAG pipelines, LLM workflows, computer vision, voicebots, chatbots, and trading infrastructure. Founder of DragonOS. Expert in Python, LangChain, PyTorch, OpenAI API, and GCP. Proven ability to own AI products 0 → 1 with no bureaucracy — shipping fast, iterating with users, and bridging deep engineering with product thinking. Actively seeking full-time remote AI roles.

WORK EXPERIENCE

AI Software Engineer · [Eden Group](#)

Mar 2025 – Present

- Built end-to-end LLM-powered AI workflows for real estate marketing and financial analysis using Python, LangChain, and OpenAI API
- Designed and deployed production RAG systems for document understanding, intelligent retrieval, and automated document management
- Developed computer vision pipelines for property image analysis and automated visual content generation at scale
- Shipped conversational chatbots and voicebots with sub-2s response latency for real-time client engagement
- Engineered ML-driven trading infrastructure integrating financial forecasting models and live market signal generation using PyTorch and Pandas
- Built OpenClaw-based agentic workflows for automated data extraction, processing, and structured output pipelines
- Architected cloud-native AI systems on GCP with scalable networking, Snowflake data pipelines, and Docker containerization
- Implemented prompt engineering best practices, evaluation frameworks, and hallucination mitigation strategies across all AI products

AI Software Engineer · [KosmosOne PVT Ltd](#)

Feb 2024 – Dec 2024

- Led a team of 5 engineers in the development of a 7-DOF Robotic Arm for autonomous space docking and undocking
- Supervised and developed the AI for autonomous locomotion using computer vision and sensor fusion in Python
- Developed high-fidelity simulations and trajectory analyses for satellite data including fuel cost and downtime metrics
- Developed a prototype for an autonomous indoor drone using SLAM-based navigation in Python
- Built an AI model for text-to-speech and sentiment analysis using multiple NLP models in Python

AI & Aerospace Software Engineer · [Freelancing](#)

Sept 2023 – Jan 2024

- Delivered AI and aerospace engineering solutions to clients across simulation, computer vision, and data analysis domains
- Built reinforcement learning models for glider locomotion training and autonomous navigation systems
- Conducted satellite trajectory analysis and developed automation tools for aerospace research workflows

Founder & Lead Engineer · [DragonOS \(Startup\)](#)

Ongoing

- Founded and built DragonOS — a software startup — owning the full product roadmap, engineering, and go-to-market strategy
- Defined quality standards, acceptance criteria, and technical architecture for all product features end-to-end

- Managed all engineering, product decisions, and user feedback loops with no external management layers

Avionics & Control Engineer Intern · OmSpace Rocket and Exploration PVT Ltd

Jan 2021 – Mar 2021

- Devised a Field Oriented Control architecture framework using MATLAB and Simulink for gimbal control of the Rutherford Engine model
- Designed satellite guidance trajectory and simulated Low Earth Orbit of 800 km using Python and Astropy

Propulsion Intern · Vayuusastra Rocket Engines and Fabrication

Sep 2019 – Dec 2019

- Designed and built a model rocket using SolidWorks CAD; designed nozzle and combustion chamber in SolidWorks and MATLAB
- Tested solid rocket motor grain designs and performed trade-off analysis to optimise thrust curves for maximum altitude

EDUCATION

MSc, Astronautics & Space Engineering — Cranfield University (*incomplete*)

Sept 2022 – Sept 2023

B.Tech, Aerospace Engineering — SRM Institute of Science & Technology

Jul 2018 – Jul 2022 · CGPA
8.12/10

Key Academic Projects:

- NEOMIR Group Project (Cranfield) — Comparative study of ESA mission vs. 6-satellite constellation in Distant Retrograde Orbit; led propulsion team, computed L1 Halo Orbit trajectory in Python and validated in STK
- Major Project (SRM) — Designed and analysed a Pintle Injector with Compressed Flow Area and Baffles using SolidWorks CAD and ANSYS Fluent
- Minor Project (SRM) — Developed a Space Debris Collision Avoidance Algorithm in Python to calculate collision probability in close proximity

TECHNICAL SKILLS

AI / ML:	LLMs, RAG Systems, Computer Vision, Agentic Workflows, Prompt Engineering, Reinforcement Learning, Fine-tuning, Neural Networks, NLP, Sentiment Analysis, SLAM
Frameworks:	LangChain, PyTorch, Keras, TensorFlow, OpenAI API, NumPy, Pandas, HuggingFace, Astropy, OpenCV
Cloud / Infra:	GCP, AWS, Docker, Snowflake, REST APIs, Git, CI/CD
Languages:	Python, C++, C, SQL, MATLAB, TypeScript, JavaScript, HTML, CSS
Aerospace:	Simulink, ANSYS Fluent, SolidWorks, CATIA, AutoCAD, STK, GMAT, Astrodynamics, AOCS, Trajectory Analysis
Domains:	Real Estate AI, Fintech & Trading, Document Intelligence, Conversational AI, Space Systems, Robotics
Product:	0 → 1 Product Development, Roadmap Ownership, Technical Specifications, Evaluation Frameworks

CERTIFICATIONS

- Neural Networks and Deep Learning — Coursera / DeepLearning.AI (Mar 2024)
- IBM Machine Learning with Python — IBM / Coursera (Dec 2024)
- Python for Everybody — University of Michigan / Coursera
- Introduction to Crash Analysis — ANSYS (May 2020)

EXTRACURRICULAR & VOLUNTEER WORK

Structural Design Engineer & Analyst · Team Spars / SRM University

Jan 2019 – Apr 2020

- Built and analysed high lift-to-weight ratio RC aircraft for national and international competitions
- Led design team to 2nd place in SAE Aeromodelling Competition 2020 (micro class) and 2nd in Boeing National Aeromodelling Competition
- Designed and manufactured test rig to study wing bending moments; performed bending and stress analysis using ANSYS and SolidWorks

Software Engineer (Volunteer) · UKSEDS / Cranfield University

Jan 2023 – Apr 2023

- Worked on autonomous navigation software for UKSEDS Robotics Competitions; studied RRT and RRT* algorithms in MATLAB
- Built a Reinforcement Learning model for rover locomotion for the rover challenge

ACHIEVEMENTS & LANGUAGES

Achievements: State Level Table Tennis Player (3rd, Club Competition) · Swimming Competition Gold Medal · 2nd Boeing National Aeromodelling · 3rd SAE International Design Report

Languages: English (Native) · Bengali (Native) · Hindi (Professional) · Japanese (Conversational)

Interests: Cooking & Gastronomy · Non-fiction Reading · Origami · Table Tennis · Swimming · Language Learning